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Saveetha Institute of Medical And Technical Sciences
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A Real-time Stock Market Simulation and Intelligent Guidance System for Beginner Investors

DSA0103- OBJECT ORIENTED PROGRAMMING WITH C++

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ABSTRACT

- The application provides a real-time stock market simulation where beginners can practice trading using virtual money in a safe environment.
- It enables users to buy and sell stocks, manage portfolios, and monitor investment performance through interactive dashboards.
- The application offers AI-based investment guidance, market insights, voice assistance, and multilingual support for an enhanced user experience.
- Gamification features such as badges, rewards, leaderboards, and daily challenges encourage continuous learning and user engagement.
- The application provides an interactive and user-friendly platform that helps beginners gain practical trading experience and make informed investment decisions.



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Engineer to Excel

INTRODUCTION



- The stock market is an important platform for investment, but many beginners find trading difficult due to limited knowledge and market risks.
- This application provides a real-time stock market simulation where users can practice trading with virtual money in a safe environment.
- Users can buy and sell stocks, manage portfolios, and monitor their investment performance through an interactive interface.
- The application includes AI-based investment guidance, market insights, multilingual support, voice assistance, and gamification features to enhance learning.
- It aims to improve financial literacy, build investment confidence, and help users develop better trading and decision-making skills.



OBJECTIVES

- To develop a real-time stock market simulation application for beginner investors.
- To provide a risk-free environment for practicing stock trading using virtual money.
- To enable users to buy, sell, and manage virtual investment portfolios.
- To provide AI-based investment guidance, market insights, and personalized recommendations.
- To support multilingual access and voice assistance for improved usability and accessibility.
- To increase user engagement through gamification features such as badges, rewards, and leaderboards.
- To improve financial literacy, investment knowledge, and decision-making skills through interactive learning.



EXISTING SYSTEM



- Existing stock market applications mainly focus on real-time trading and portfolio management.
- Most platforms require users to invest real money, making them unsuitable for beginner investors.
- Many applications provide limited educational content and beginner-friendly investment guidance.
- AI-based recommendations, multilingual support, and voice assistance are not commonly integrated into a single platform.
- Gamification features such as badges, rewards, leaderboards, and interactive learning are absent or limited in many existing systems.



PROPOSED SYSTEM



- The proposed application provides a real-time stock market simulation using virtual money for risk-free trading practice.
- It enables users to buy and sell stocks, manage portfolios, and track investment performance through an interactive interface.
- The system offers AI-based investment guidance, market insights, and personalized recommendations to support better decision-making.
- It includes multilingual support, voice assistance, and gamification features to improve accessibility, learning, and user engagement.
- The application helps beginners develop investment knowledge, build confidence, and gain practical trading experience in a safe environment.

LITERATURE SURVEY



Title	Author(s)	Year	Journal / Conference	Publication (Vol./Issue/Pages)	components & Techniques Used	Limitation
Deep Learning in the Stock Market: A Systematic Survey of Practice, Backtesting, and Applications	Kenniy Olorunnimbe, Herna Viktor	2022	Artificial Intelligence Review	Vol. 56, Issue 3	Brian Ferrell, Bridget Thomson McInnes	Does not support beginner learning or trading simulation.
Impact of Gamification on Mitigating Behavioral Biases of Investors	Doğaç Şenol, Ceylan Onay	2023	Journal of Behavioral and Experimental Finance	Vol. 37	and trading behavior using Gamification and Behavioral Finance	Does not provide AI guidance or multilingual support.
Trading Gamification and Investor Behavior	Philipp Chapkovski, Mariana Khapko, Marius Zoican	2024	Management Science	Vol. 70, Issue 9	Trading engagement using Gamification and Experimental Finance	Focuses on investor behavior rather than portfolio management.
Reinforcement Learning and NLP for Stock Market Trading: A Methodological Survey	Brian Ferrell, Bridget Thomson McInnes	2025	ACM Computing Surveys	Vol. 58, Issue 1	Stock trading using Reinforcement Learning (RL) and Natural Language Processing (NLP)	Focuses mainly on trading strategies and prediction.

MODULE 1: SMART STOCK MARKET SIMULATION

- Simulates real-time stock prices, market fluctuations, and trading activities to provide users with a realistic virtual stock market experience.
- Displays detailed stock information, company profiles, historical price trends, financial news, and live market updates.
- Supports multilingual interfaces, allowing users to access the application in their preferred language for better accessibility and ease of use.
- Provides an intelligent voice assistant for searching stocks, navigating application features, checking market updates, and receiving investment guidance.
- Generates realistic market events, economic scenarios, daily trading challenges, and educational activities to enhance practical learning and decision-making skills.



MODULE 2: SMART TRADING AND PORTFOLIO MANAGEMENT



- Enables users to buy and sell virtual stocks using virtual money in a completely risk-free trading environment.
- Allows users to create, manage, and monitor diversified investment portfolios and personalized stock watchlists.
- Maintains complete transaction records, trading history, and portfolio summaries for effective investment tracking.
- Automatically calculates profit, loss, return on investment (ROI), portfolio value, and overall trading performance.
- Sends smart notifications, stock price alerts, portfolio updates, and investment reminders to keep users informed about market activities.

MODULE 3: INTELLIGENT INVESTMENT SUPPORT

- Provides AI-based investment recommendations and personalized market insights based on stock performance and user preferences.
- Performs investment risk analysis, portfolio health assessment, and suggests suitable investment strategies for beginners.
- Offers beginner-friendly tutorials, interactive quizzes, personalized learning paths, and educational resources to improve financial knowledge.
- Analyzes financial news, market sentiment, and stock performance to provide clear explanations and informed investment suggestions.
- Includes gamification features such as badges, rewards, experience points (XP), achievement milestones, levels, leaderboards, and daily challenges to encourage continuous learning.



MODULE 4:ANALYTICS, REPORTS, AND INTELLIGENT INSIGHTS



- Generates comprehensive portfolio performance, trading history, investment summaries, and financial reports for detailed analysis.
- Displays interactive dashboards, charts, graphs, and market analytics to help users visualize their investment performance.
- Predicts future stock trends using AI-based analytical techniques and historical market data to support better investment decisions.
- Securely stores user profiles, portfolio details, transaction records, stock information, and application data using a MySQL database.
- Supports goal-based investment planning, personalized recommendations, intelligent investment tips, and detailed performance reports to improve long-term investment strategies.



TECHNOLOGY USED



- Programming Language: C++ for developing the application and implementing the core functionalities.
- Database: MySQL for storing user information, portfolio details, stock data, and transaction records.
- Programming Paradigm: Object-Oriented Programming (OOP) for modular, reusable, and maintainable code.
- AI Techniques: Machine Learning concepts for investment recommendations, market analysis, and stock trend prediction.
- Supporting Technologies: Voice Assistant, Multilingual Support, Data Analytics, and Gamification for an interactive and intelligent user experience.

CONCLUSION

- The proposed application provides a safe and realistic virtual stock market environment where beginners can learn and practice stock trading without investing real money.
- It enables users to buy and sell virtual stocks, manage investment portfolios, and monitor trading performance through an interactive and user-friendly interface.
- AI-based investment guidance, multilingual support, voice assistance, and gamification features improve user engagement and help users make informed investment decisions.
- Advanced analytics, portfolio reports, market insights, and personalized recommendations enhance users' financial knowledge and practical trading experience.
- Overall, the application serves as an effective learning platform that builds confidence, improves financial literacy, and prepares beginners for real-world stock market investing.

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THANK YOU